Forex Price Alert System

Phase 1 Requirements | Web dashboard + Telegram alerts | Target: about 7 working days

1. Project Summary

A **web-based Forex price alert system**. The user enters support and resistance levels for multiple currency pairs. The system watches live prices and sends an **instant Telegram notification** when price is approaching or touching a level. The user then decides manually whether to trade.

The system only sends alerts. It does not trade and gives no financial advice. Strategy logic, MT5 and trade execution are later phases (see Section 6).

Item	Detail
Product type	Responsive web app (not a mobile app)
Core function	Price-level alerts for Forex pairs
Notification channel	Telegram bot
Pairs	Multiple Forex pairs
Levels	Customizable support and resistance levels
Alert content	Pair, current price, level, and status: approaching or touching
Trading	None in this phase (no automatic entries)
Deliverables	Working web app, full source code, short usage guide

2. Milestones

Milestone	Scope	Target
1. Dashboard and level setup	Responsive dashboard; pair management (add, remove, enable/disable); add, edit, delete support and resistance levels; clean levels view; architecture ready for live data and Telegram	Days 1-3
2. Live price monitoring	Live Forex price source; live prices on dashboard; background worker comparing price to every active level; approaching/touching detection; reconnect and error handling	Days 4-5
3. Telegram alerts and testing	Telegram bot and account linking; instant alerts; alert history in dashboard; end-to-end testing and bug fixing; deployment and handover	Days 6-7

3. Functional Requirements

3.1 Dashboard and levels

ID	Requirement
FR-1	User can log in securely (assumption: email and password).
FR-2	User selects from a list of Forex pairs: majors (EURUSD, GBPUSD, USDJPY, AUDUSD, USDCAD, USDCHF, NZDUSD), plus XAUUSD if the data source supports it.
FR-3	For each pair the user can create, edit, delete and enable/disable levels.

ID	Requirement
FR-4	Each level has: pair, price, type (support or resistance), optional note, active toggle.
FR-5	Dashboard shows each pair with live price, its levels, and distance to the nearest level.
FR-6	Dashboard shows status per level: Idle, Approaching, Touched.
FR-7	Alert history page lists past alerts with time, pair, level, price and status.

3.2 Price monitoring and detection

ID	Requirement
FR-8	System receives live prices for all pairs that have active levels.
FR-9	**Approaching**: price is within a configurable distance of a level (default 10 pips).
FR-10	**Touching**: price is within a small tolerance (default 1-2 pips) or crosses the level.
FR-11	Pip size handled correctly: 0.0001 for most pairs, 0.01 for JPY pairs.
FR-12	Anti-spam: one approaching and one touching alert per level; reset only after price moves away by a reset distance or after a cooldown (default 30 minutes).
FR-13	Distance, tolerance and cooldown are configurable in settings.

3.3 Telegram notifications

ID	Requirement
FR-14	A Telegram bot sends messages to the user's chat.
FR-15	User links Telegram by sending /start with a code shown in the dashboard.
FR-16	Alert message includes pair, status, level, current price and time (example below).
FR-17	A "Send test alert" button confirms the connection works.
FR-18	If delivery fails, the system retries and logs the failure.

Example alert:
Price Alert: EURUSD
Status: Touching Resistance
Level: 1.08500 | Current price: 1.08498
Time: 10 Nov 2026, 14:32 UTC

4. Non-Functional Requirements

- **Responsive:** works on desktop and mobile browsers.
- **Reliability:** monitoring runs 24/5 without manual restarts; auto-reconnect on feed drops.
- **Speed:** alert delivered within a few seconds of the condition being met (depends on data source).
- **Security:** hashed passwords; API keys and bot token in environment variables only.
- **Maintainability:** clean structure, documented setup, easy to extend later.
- **UI style:** dark, trading-style dashboard following the client's prototype look.

5. Suggested Technical Approach

Layer	Suggestion
Front end	React or Next.js, Tailwind CSS

Layer	Suggestion
Back end	Node.js (Express) or Python (FastAPI)
Database	PostgreSQL (or Supabase): users, pairs, levels, alerts
Monitoring worker	Separate background process that streams or polls prices and evaluates levels
Telegram	Telegram Bot API (webhook or long polling)
Price data	Twelve Data, OANDA, Finnhub or similar; choose one with websocket or fast polling
Hosting	VPS or Render/Railway (worker must run continuously)

Design note: keep the price-data layer behind a small interface so MT5 can be added later as another price source and for trade execution. Check free-tier API limits against the number of pairs before choosing a provider.

6. Out of Scope for Phase 1

- Bollinger Bands (200, deviation 3) and 25/50 SMA crossover logic
- Daily-candle consolidation boxes and wick analysis
- 5-minute breakout confirmation and 15-minute monitoring
- Automatic support and resistance grid generation
- Projected trade path display
- MT5 connection, one-click trade execution, stop-loss and take-profit management
- Mobile app, Pine Script version, multi-user distribution and subscriptions

7. Acceptance Criteria

Milestone	Accepted when
1	User can log in, add pairs, create, edit and delete levels, and view them on a responsive dashboard.
2	Live prices update for at least 5 pairs, and approaching/touching statuses change correctly in a test.
3	Correct Telegram alerts arrive for a test level, with no duplicate spam; history is logged; source code and guide are delivered.

8. Open Questions

#	Question
1	Single user or multiple users? (The client once mentioned that "people can download the bot".)
2	Third-party price feed acceptable? Prices may differ slightly from the client's MT5 broker feed.
3	Exact pairs list; are gold or crypto needed?
4	How many pips count as "approaching"?
5	Can the client provide a few real levels for testing?
6	Who creates the Telegram bot (BotFather) and holds the bot token?
7	Which server and data API plan will be used (is a paid tier needed)?
8	Review the client's prototype (Target_AI_Trader_iPhone_App.zip and screenshots) for the look and layout to match.